Summary in Graph

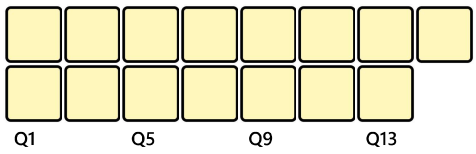
Exam Summary (GO Classes DA Test Series 2025 | Artificial Intelligence | Topic wise Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (51) out of 505 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Select the statement that is the **negation** of "Some weasels are cuddly."

A. No weasels are cuddly.
B. All weasels are cuddly.
C. Some weasels are not cuddly.
D. All cuttlefish are weasely.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 04sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

We use the dictionary:

M	domain of men
s	Sharon
$N(x)$	x is nice
$L(x, y)$	x loves y

Which of the following formulas corresponds to the sentence:

$$\text{There is a nice man who loves Sharon.}$$

A. $\exists x \in M[N(x) \wedge L(x, s)]$
B. $\exists x \in M[N(x) \rightarrow L(x, s)]$
C. $\forall x \in M[N(x) \wedge L(x, s)]$
D. $\forall x \in M[N(x) \rightarrow L(x, s)]$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider the following two logical equivalences that express 'logical laws':

- $\neg\forall_{\mathrm{x}\in\mathrm{D}}\mathrm{f} \equiv \exists_{\mathrm{x}\in\mathrm{D}}\neg\mathrm{f}$
- $\neg(f \leftrightarrow g) \equiv \neg f \leftrightarrow \neg g$

Which of these hold?

A. both
B. only the first
C. only the second
D. none

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider the sentence:
Nadhi eats a fruit only if the fruit is an apple.
Define the propositional variables:
n : Nadhi eats a fruit.
a : The fruit is an apple.
Which of the following is a translation of this sentence into propositional logic?

A. $(n \rightarrow a)$
B. $(a \rightarrow n)$
C. $(n \leftrightarrow a)$
D. $(n \wedge a)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Let $\mathrm{Bird}(x)$ means " x is a bird" and $\mathrm{Fly(y)}$ means " y can fly". Which of the following is a correct logical statement for "No bird can fly"?

A. $\forall x(\sim \mathrm{Fly}(\mathrm{Bird}(x)))$
B. $\forall x(\mathrm{Bird}(x) \wedge \sim \mathrm{Fly(x)})$
C. $\forall x(\mathrm{Bird}(x) \rightarrow \sim \mathrm{Fly(x)})$
D. $\forall x \forall y(\mathrm{Bird}(x) \rightarrow \sim \mathrm{Fly(y)})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Consider the following interpretation of a language with a unary predicates P, Q, and a binary predicate R.

- Universe $= \{1, 2, 3, 4\}$
- $P = \{\langle 1\rangle, \langle 3\rangle\}$
- $Q = \{\langle 2\rangle, \langle 4\rangle\}$
- $R = \{\langle 3, 2\rangle, \langle 4, 3\rangle, \langle 3, 1\rangle, \langle 4, 2\rangle, \langle 2, 1\rangle, \langle 4, 1\rangle\}$

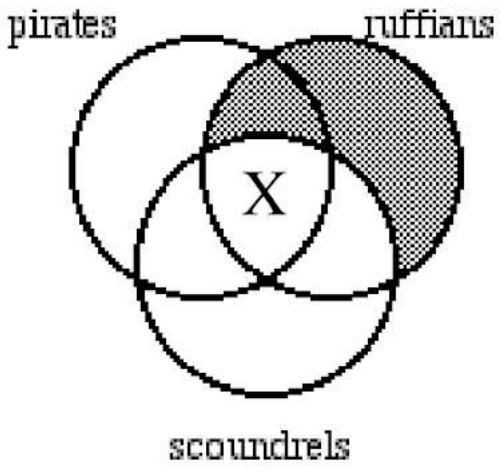
Which of the following sentences below hold in this interpretation?

A. $\exists x(P(x) \wedge Q(x))$
B. $\exists x\,(P(x) \rightarrow Q(x))$
C. $\forall x\,(P(x) \rightarrow \exists y(R(x, y)))$
D. $\forall x\,(Q(x) \rightarrow \exists y(R(x, y)))$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Suppose the marked diagram below conveys information about relationships between pirates, ruffians and scoundrels. We use shading to indicate that a region contains no elements. An " X " in a region indicates the existence of at least one element; an " X " on the boundary between two regions indicates that the union of those two regions contains at least one element. If a region is unmarked, then whether that region contains any elements is uncertain. Select the choice that must be true according to the marked diagram.



A. No scoundrels are ruffians and some pirates aren't scoundrels.
B. All ruffians are scoundrels and some scoundrels aren't pirates.
C. No pirates are scoundrels and some ruffians are scoundrels.
D. All ruffians are scoundrels and some pirates are ruffians.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

The statement form $(p \Leftrightarrow r) \Rightarrow (q \Leftrightarrow r)$ is equivalent to

A. $[(\sim p \vee r) \wedge (p \vee \sim r)] \vee \sim [(\sim q \vee r) \wedge (q \vee \sim r)]$
B. $\sim [(\sim p \vee r) \wedge (p \vee \sim r)] \wedge [(\sim q \vee r) \wedge (q \vee \sim r)]$
C. $[(\sim p \vee r) \wedge (p \vee \sim r)] \wedge [(\sim q \vee r) \wedge (q \vee \sim r)]$
D. $\sim [(\sim p \vee r) \wedge (p \vee \sim r)] \vee [(\sim q \vee r) \wedge (q \vee \sim r)]$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Which of the following formulas is a formalization of the sentence:
"Every connected and circuit-free graph is a tree"
Where $\mathrm{cir(x)}$ means " x has circuit".

A. $\forall \mathrm{x}[(\mathrm{conn(x)} \wedge \neg\, \mathrm{cir(x)}) \rightarrow \mathrm{tree(x)}]$
B. $\neg \exists \mathrm{x}[(\mathrm{conn(x)} \wedge \neg\, \mathrm{cir(x)}) \wedge \neg\, \mathrm{tree(x)}]$
C. $\forall \mathrm{x}[(\mathrm{conn(x)} \wedge \neg\, \mathrm{cir(x)}) \wedge \mathrm{tree(x)}]$
D. $\forall \mathrm{x}[(\mathrm{conn(x)} \wedge \mathrm{cir(x)}) \rightarrow \mathrm{tree(x)}]$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Given the statement form below where p, q, r, s are statement variables,

$$\Big(\Big(\big((p \rightarrow q) \rightarrow (q \rightarrow r)\big) \rightarrow (r \rightarrow s)\Big) \vee p\Big) \wedge (p \rightarrow (p \rightarrow q)) \wedge p$$

Which of the following is logically equivalent to the above?

A. p
B. $p \vee q$
C. $p \wedge q$
D. $p \rightarrow q$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

The island of Wantuutrewan is inhabited by exactly two types of natives: knights who always tell the truth and knaves who always lie. Every native is a knight or a knave, but not both.
You meet two islanders and here is what they say:

- Aiken says: Dueet and I are the same type of people.
- Dueet says: Aiken and I are different types of people.

Which of the following is true?

A. Both Aiken and Dueet are knights.
B. Aiken is a knight and Dueet is a knave.
C. Aiken is a knave and Dueet is a knight.
D. Both Aiken and Dueet are knaves.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Given the following where p, q, r, s, t are statement variables:
(i) $(\sim r \rightarrow \sim r) \vee (\sim q \rightarrow q) \vee (t \rightarrow \sim t)$
(ii) $(p \wedge q \wedge s) \vee (\sim p \wedge q \wedge \sim r \wedge t) \vee q \vee (q \wedge r \wedge s \wedge \sim t)$
(iii) $((t \wedge s) \rightarrow (t \vee s)) \wedge (q \rightarrow \sim q)$
(iv) $(\sim q \rightarrow q) \wedge (r \rightarrow r)$
Which of the above are logically equivalent to one another?

A. Only (i) and (iii).
B. Only (ii) and (iv).
C. Only (i), (ii) and (iv).
D. Only (ii), (iii) and (iv).

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Assume the domain is $\mathbb{R}$. Which of the following statements are true?

i. $\forall x \exists y (xy = 1)$
ii. $\exists x \forall y \left(x^2 \geq y^2\right)$
iii. $\exists x \forall y \exists z \left(xy^2 = yz\right)$

A. Only (ii).
B. Only (iii).
C. Only (i) and (ii).
D. Only (i) and (iii).

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Let S be the set of students and $x \in S$. We define the following predicates:
$\mathrm{SoC}(x)$: x is a computing student;
$\mathrm{CS\,1231}(x)$: x has taken CS1231S.
Given that the statement $\forall x \in S(\mathrm{SoC}(x) \wedge \mathrm{CS1231}(x))$ is true, which of the following statements are true?

i. All computing students have taken CS1231S.
ii. Every student is a computing student.
iii. There must be at least one student in the set S.
iv. Some non-Computing student has taken CS1231S.

A. Only (i).
B. Only (i) and (ii).
C. Only (ii) and (iv).
D. Only (i), (ii) and (iv).

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Which of the following statements are true?

i. $\sim (p \rightarrow q) \equiv (p \vee (p \rightarrow \sim q))$
ii. $(p \rightarrow (q \vee r)) \leftrightarrow ((p \wedge \sim q) \rightarrow r)$
iii. $((p \leftrightarrow q) \wedge (q \rightarrow r) \wedge (r \vee \sim s) \wedge (\sim s \rightarrow q)) \rightarrow s$

A. Only (ii).
B. Only (i) and (ii).
C. Only (i) and (iii).
D. Only (ii) and (iii).

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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